



Al-Balqa Applied University

Faculty of Engineering Technology, Basic Science Department, MID Exam

Statistics and prob. for eng. / Prob. and rand. processes

Instructor name:

Student Name:

Lecture Time:

Question Number	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Answer	B	C	D	A	B	D	D	D	A	C
Question Number	Q11	Q12	Q13	Q14	Q15					
Answer	✓	✓	X	✓	✓					

Question One: If $P(A|B) + P(B|A) = 0.8$ and

$$\frac{1}{P(B)} + \frac{1}{P(A)} = 4$$

Then $P(B \cap A) =$

- A) 0.1 B) 0.2 C) 0.3 D) 0.4 E) None

Question Two: For each positive integer n , we have $\binom{n}{0} =$

- A) 0 B) $n!$ C) 1 D) undefined E) None

Question Three: A discrete random variable X with values $0, 1, -1$ with $P(X=1) = P(X=-1) = c$. If $\text{Var}(X) = 4$; then $c =$

- A) 0 B) 1 C) 3 D) 2 E) None

Question Four: If $P(A - \bar{B}) + P(A - B) = 0.1$, then $P(A) =$

- A) 0.1 B) 0.9 C) 0.2 D) 0.8 E) None

Question Five: Let f be a continuous function with $f(0) = 0$. If $\int_{-\infty}^{\infty} (2f(x-1) + f(x+1)) \delta(x+1) dx = 4$; then $f(-2) =$

- A) 1 B) 2 C) 3 D) 4 E) None

Question Six: Let X be a random variable with $E(X) = 1$ and $E(X^2) = 2$. We have $E(X^2 + X + \text{Var}(X)) =$

- A) 1 B) 2 C) 3 D) 4 E) None

Question Seven: For continuous random variable X with cdf $F(x)$. If the cdf $F(x)$ is constant on the interval $[0, 5]$ then find $P(0 < x < 2)$

- A) 0 B) 1 C) 3 D) 2 E) None

Question Eight: Let X, Y be two independent random variables. If $\text{Var}(X) = 4$ and

$$(E(Y))^2 E(X^2) - (E(XY))^2 = 16$$

Then $|E(Y)| =$

- A) 0.25 B) 1 C) 3 D) 2 E) None

Question Nine: For the joint PDF $f(x, y) = b^b$ when $0 < x < 1$, $0 < y < b^{-2}$ and zero otherwise, the positive constant $b > 1$ equals

- A) 2 B) 4 C) 3 D) 5 E) None

Question Ten: Two independent events A, B with equal probability; if $P(A \cup B) = 0.5$; then $P(A) =$

- A) $1 + \frac{1}{\sqrt{2}}$ B) $1 - \frac{1}{\sqrt{2}}$ C) $1 \pm \frac{1}{\sqrt{2}}$ D) $\sqrt{2}$
E) None

Question Eleven ($\checkmark$, $\times$): A discrete random variable X with values $1, -1, 0$. We must have $E(X^2) = E(X^4)$ ✓

Question Twelve ($\checkmark$, $\times$): For a random variable X with cdf $F(x) = 1 - 0.5e^{-2x}$, $x > -0.5 \ln(2)$, the pdf has to be $f(x) = e^{-2x}$ ✓

Question Thirteen ($\checkmark$, $\times$): When x is nonzero, we have $u(x) + u(-x) = 0$ $\times$

Question Fourteen ($\checkmark$, $\times$): If $\text{Cov}(-X, -Y) = \sigma_X$ then $\sigma_Y > 1$ $\checkmark$

Question Fifteen ($\checkmark$, $\times$): For the points of $R_{XY} = \{(a, a), (a, b), (b, a), (b, b)\}$ $\checkmark$
assume that $f(a, a) = f(b, b) = \delta$ and $f(a, b) = f(b, a) = 2\delta$. Then

$$\delta = \frac{1}{4}$$